

A 22nm 96KX144 RRAM macro with a self-tracking reference and a low ripple charge pump to achieve a configurable read window and a wide operating voltage range

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Abstract

An RRAM macro equips a hybrid self-tracking reference and a low ripple charge pump is presented. It realizes configurable read windows and a consistent write performance at operation voltage range of 1.62V~3.63V. It shows 6.5ns and 10ns access time at 0.7V can be achieved for OTP (one-time-program) and 10K endurance applications, respectively.

Introduction

RRAM is a promising non-volatile memory technology for embedded applications due to its low cost and compatibility to logic process. The reliability of RRAM is strongly relating to its resistance state, the high resistance state (HRS) or the low resistance state (LRS). Write-verify process is commonly used to achieve the target resistance states [1-3]. Through the read SA to do verification can ensure the read window to be established. A constant current source is used for read/verify in the prior art [1]. However, verifying with a constant current source at HT (125C)/CT (-40C) would over-/under-write RRAM element (RE) and impact reliability. Some would use a mini-array to track temperature and process variation [2]. However, RE resistance will drift over time and hard to be a reliable reference. In addition, the parasitic IR-drop along the write path will influence the write efficiency at different temperature. To solve above issues, we adopt a proportional to absolute temperature (PTAT) write bias to compensate the IR-drop and present a self-tracking reference current to achieve target resistance states [4].

Low-Ripple Charge Pump Scheme

Fig.1 depicts a 13.5Mb RRAM macro that comprises 28 blocks. Each has 640X768 cells sharing a local LDO to generate a PTAT and adjustable write bias VBL. The power supply of the LDOs can be VDIO or from a low-ripple charge pump (LR-CP) steered by a VDIO-detection (VDIO-DET). LR-CP will be enabled as VDIO < 2.5V. The conventional switched capacitor charge pump is hard to achieve a low output ripple [5]. However, a big output ripple may impact the stability of RRAM write performance. To realize a low output ripple usually requires a big reservoir capacitance [5-6]. A presented scheme depicted in Fig 2 is to solve the issue. The salient invention is the load-aware detection circuit (LA-DC) which can adaptively deploy the required pumps according to the current load. At a light load condition, one pump is active to keep VPMP to the maximum, V1. As current load exceeds one pump's driving ability, VPMP will descend and trigger more pumps sequentially. Ultimately, the four active pumps should cover the heaviest load. LA-DC is implemented as a voltage comparator with multiple reference levels. Node COM is adaptively adjusted to redistribute the branch currents. LA-DC can convert ripples to a lower frequency region, so that the output ripple can be further suppressed by the power-supply-rejection of the VBL LDO. The performance of LA-DC compared with the

conventional one level detection is checked and shown in Fig.3. With the scheme, a 30mV of output ripple amplitude of VBL is measured as Fig.4. A consistent write performance is checked by performing one-shot SET and RESET from 1.62V to 3.63V. LR-CP provides a stable power supply and only requires a 20pF of reservoir capacitor.

The Hybrid Self-Tracking Reference

Fig.5 depicts a 640X768 block that consists of 10 subarrays. Each subarray has a read-SA sitting between two 32X768 cell arrays. The column multiplexer is adopted to support bipolar write [1-4]. BMUX is to sink current and cascaded with a pull down device to act as a current limiter for forming and SET [4]. RESET is done by digital drivers. When a read asserts, the SA will be connected to the selected cell via SC_L/SC_R. The associated switch SR_R/ SR_L will connect the unselected side to the reference branch to cancel out the off current. The SA operation waveform is shown in Fig.6. The presented self-tracking reference 4T3R network is shown as Fig.7. The 4T comprises of 2 shunted NMOS-pairs connected in series to emulate the select-transistor and reflect the temperature effect. 3R is a thin-film resistor network to emulate three targeting resistance states, the maximum LRS (RL), the minimum HRS (RH) and the reference level for normal read (R_NR). The read window (ΔI) is the difference between Ir0 and Ir1 which corresponds to the cell current at RH and RL. A temperature-invariant current (TIC) is shunted to 4T3R through the current mirror CM. The TIC is used to adjust the resultant reference current level if the thin-film resistance deviates from the target. Fig.8 shows the cell currents read at RT (25C) after written at HT/RT/CT respectively. The overlap of 3 Ir0/Ir1 distributions proves the 4T3R reference scheme can achieve the same resistance state when write is done at different temperatures.

Silicon Measurement Results

The retention-after-cycling (RAC) test is to bake samples at 175C for 11 hours after 10K times of SET+RESET operations. Fig.9 shows a 7uA of read window is achieved after 10K RAC and a 28uA of read window is achieved after OTP RAC. Fig. 10 shows a 10ns of access time (Tacc) at 0.7V is measured after 10K RAC and a 6.5ns of Tacc is achieved for OTP's. The presented 4T3R reference demonstrates a configurable read window and achieves the consistent resistance states as write is performed at different temperatures. Fig.11 is the design summary and Fig.12 is the die photograph.

References

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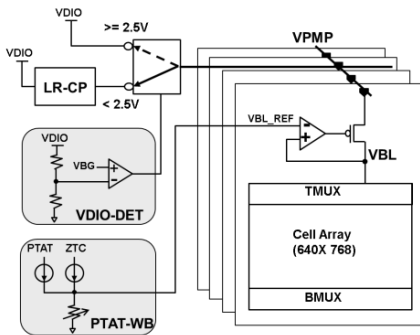


Fig.1 13.5Mb macro architecture

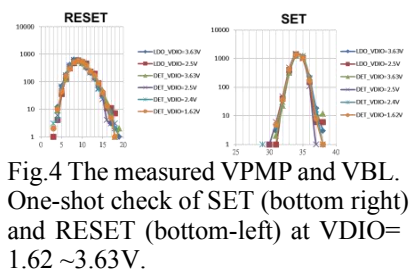
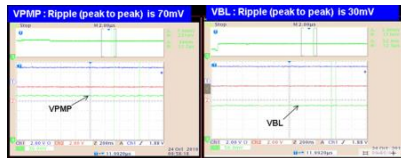


Fig.4 The measured VPMP and VBL. One-shot check of SET (bottom right) and RESET (bottom-left) at $V_{DI0} = 1.62 \sim 3.63V$.

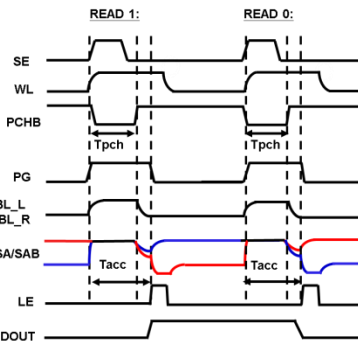


Fig.7 Waveforms of read operation

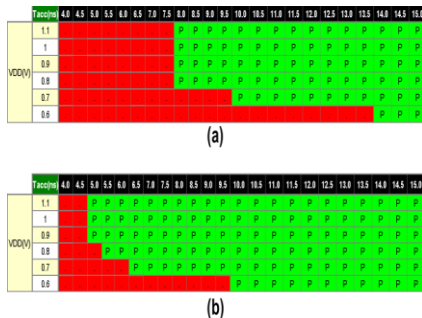


Fig.10 The shmoo of T_{acc} v.s. V_{DD} for: (a) 10KC (b). OTP operation.

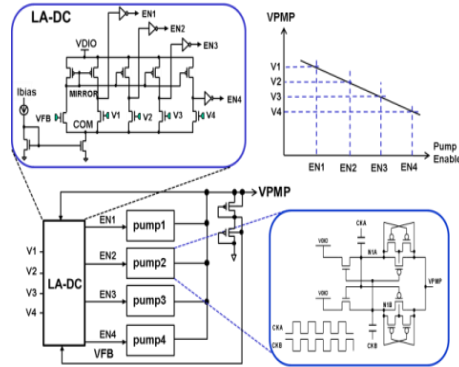


Fig.2 Low Ripple Charge Pump (LR-CP) scheme

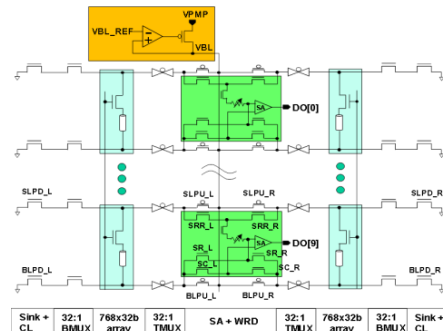


Fig.5 The architecture of 640X768 sub-block.

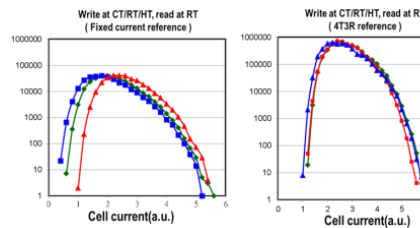


Fig.8 The cell current distribution check is done at RT after written at HT, RT and CT, respectively.

Technology node	22nm CMOS logic
Supply	0.8V / 1.62~3.63V
Cell size	53F ²
Architecture	96KX144
Density (excluding analog)	10.24Mb/mm ²
Write bias, VBL (V)	1.5~2.2V +/-20mV
Read speed @0.7V	6.5ns (OTP) 10ns (10KC)
Read window after RAC	28uA (OTP) 7uA (10KC)

Fig.11 Design summary

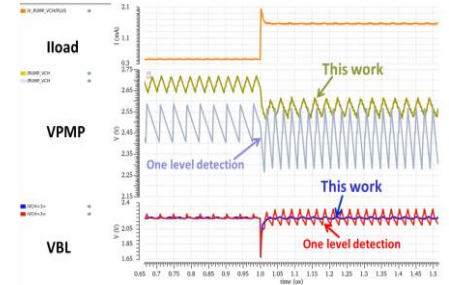


Fig.3 The transient comparison between with and without LA-DC.

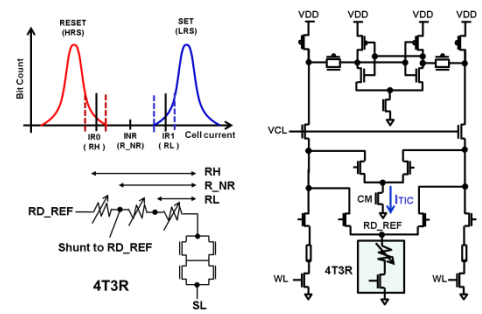


Fig.6 Read SA with self-tracking reference and step current

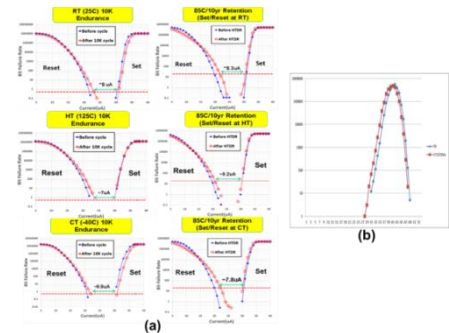
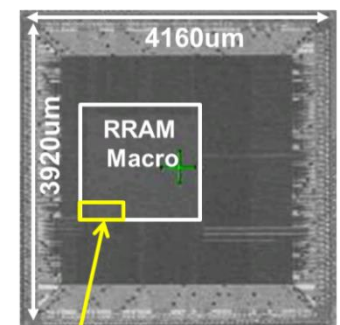


Fig.9 Read window measurement: (a) 10KC, (b) OTP



LR-CP (0.07mm²)
20pF decap included

Fig.12 Die microphotograph